A First Course in Abstract Algebra

Section 49: The Isomorphism Extension Theorem

Detailed worked solutions

Mathjock

Text and numbering. John B. Fraleigh, *A First Course in Abstract Algebra*, seventh edition (2003), following the scanned textbook linked on Mathjock. A label such as 12.3 means Exercise 3 in Section 12; A.3 means Exercise 3 in the matrix appendix. Section 0 is included. These are independently written solutions; brief descriptions identify the exercises, and the textbook supplies their original statements.

How to read the solutions. Identify the assumptions and the conclusion before beginning a proof. Keep track of the underlying set, its operations, and the hypotheses of each theorem. For a calculation, follow the intermediate steps and check the result in the original definition. A counterexample must satisfy all assumptions and fail the proposed conclusion.

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49 The Isomorphism Extension Theorem

Exercise 49.1: Extending an identity on two radicals

An embedding of E is a sign choice on $\sqrt{2}, \sqrt{3}, \sqrt{5}$. Fixing $\sqrt{2}$ forces its sign to be positive; fixing $\sqrt{15} = \sqrt{3}\sqrt{5}$ forces the other two signs to agree. Thus the two triples of images are $(\sqrt{2}, \sqrt{3}, \sqrt{5})$ and $(\sqrt{2}, -\sqrt{3}, -\sqrt{5})$. Each is realized by a sign-change automorphism, and there are no others.

Exercise 49.2: Extending a prescribed negative product

Here $\sqrt{2}$ must stay fixed and the product $\sqrt{3}\sqrt{5}$ must change sign. Thus the two triples of images are $(\sqrt{2}, -\sqrt{3}, \sqrt{5})$ and $(\sqrt{2}, \sqrt{3}, -\sqrt{5})$. They both respect the prescribed map, and exhaust its possible extensions.

Exercise 49.3: Extending conjugation on a triple product

The product of the three radical signs must be negative. The four triples are

$$(-\sqrt{2}, \sqrt{3}, \sqrt{5}),\ (\sqrt{2}, -\sqrt{3}, \sqrt{5}),\ (\sqrt{2}, \sqrt{3}, -\sqrt{5}),\ (-\sqrt{2}, -\sqrt{3}, -\sqrt{5}).$$

Every triple defines a sign-change automorphism of E, and exactly the odd numbers of sign changes negate $\sqrt{30}$.

Exercise 49.4: Embeddings of a real cubic field

Put $r = \sqrt[3]{2}$ and $\omega = (-1 + i\sqrt{3})/2$. The polynomial $x^3 - 2$ is irreducible by Eisenstein and has distinct roots $r, r\omega, r\omega^2$. For each $j = 0, 1, 2$, there is one rational embedding sending r to $r\omega^j$. Its value on $a + br + cr^2$ is $a + br\omega^j + cr^2\omega^{2j}$. These are all three embeddings into $\overline{\mathbb{Q}}$.

Exercise 49.5: Adding an independent square root

Let r, ω be as in Exercise 49.4. The field $\mathbb{Q}(r)$ has degree 3 and cannot contain the quadratic field $\mathbb{Q}(\sqrt{3})$, by the tower law. Thus adjoining $\sqrt{3}$ doubles the degree. For each of the three choices $r \mapsto r\omega^j$, the extension theorem allows both roots $\sqrt{3} \mapsto \pm\sqrt{3}$ of its transported quadratic minimal polynomial. These six independent pairs describe all embeddings; a field map is determined by these generators.

Exercise 49.6: Extending a prescribed quadratic conjugation

The source field is $\mathbb{Q}(i, \sqrt{3}, r)$ and the map must send $\sqrt{3}$ to $-\sqrt{3}$. We can independently send i to $\pm i$ and r to $r\omega^j$ for $j = 0, 1, 2$. To justify existence, $\mathbb{Q}(i, \sqrt{3})$ is quadratic over $\mathbb{Q}(\sqrt{3})$, and $x^3 - 2$ remains irreducible over it: if a cubic root lay in this degree-four field, its degree-three rational subfield would violate the tower law. Thus there are exactly $2 \cdot 3 = 6$ extensions. For each, values on the three generators determine the entire map.

Exercise 49.7: A transcendental sign change

(a) The fixed field is $\mathbb{Q}(\pi^2)$. Indeed, $\mathbb{Q}(\pi)$ has basis $1, \pi$ over $\mathbb{Q}(\pi^2)$ because π is transcendental and $x^2 - \pi^2$ gives degree 2. The sign change fixes $A + B\pi$ precisely when $B = 0$.

(b) The element $\sqrt{\pi}$ satisfies $x^2 - \pi$ over $\mathbb{Q}(\pi)$. Transporting coefficients under $\pi \mapsto -\pi$ gives $x^2 + \pi$, whose roots in the algebraic closure are $\pm i\sqrt{\pi}$. The two extensions therefore send $\sqrt{\pi}$ to these two values. They exist by the extension theorem and are determined by this image.

Exercise 49.8: Ten statements about extensions of isomorphisms

(a) False. Take $F = \mathbb{Q}$ and α transcendental; no embedding of $\mathbb{Q}(\alpha)$ into the algebraic closure $\overline{\mathbb{Q}}$ can send a transcendental element to an algebraic one.

(b) True, by the isomorphism extension theorem for algebraic extensions.

(c) False. The identity on $\mathbb{Q}$ has one extension to $\mathbb{Q}$ but two to $\mathbb{Q}(\sqrt{2})$.

(d) True. An isomorphism extends to an isomorphism of algebraic closures.

(e) False. $\mathbb{Q}$ and $\mathbb{Q}(\sqrt{2})$ are not isomorphic, but share an algebraic closure up to isomorphism.

(f) True. Both closures are algebraic closures of $\mathbb{Q}$.

(g) True. The number of embeddings is at most the finite degree.

(h) True. Count extensions of each lower embedding through a tower; each has the same number of extensions as the relative index.

(i) True. Successively extending across finitely many algebraic generators constructs the map; the infinite case needs the maximality argument.

(j) False. $\mathbb{C}$ is uncountable, whereas $\overline{\mathbb{Q}}$ is countable, so they cannot be isomorphic.

Exercise 49.9: An algebraic self-embedding of a closed field

The image $\sigma[K]$ is algebraically closed because it is isomorphic to the algebraically closed field K. By assumption every element of K is algebraic over this image. An algebraically closed field has no proper algebraic extension: each minimal polynomial is linear. Therefore every element of K already lies in $\sigma[K]$. Thus the embedding is onto and is an automorphism.

Exercise 49.10: Extending to an automorphism of the closure

Let $\sigma : E \to \overline{F}$ fix F. The extension theorem extends it to an embedding $\tau : \overline{F} \to \overline{F}$. Its image contains F, and $\overline{F}$ is algebraic over F, hence algebraic over the larger image $\tau[\overline{F}]$. Exercise 49.9 now makes τ onto. Its restriction to E is the desired original map.

Exercise 49.11: Closures of algebraically related fields

An algebraic closure $\overline{E}$ of the algebraic extension E/F is algebraic over F by transitivity of algebraicity. It is also algebraically closed. Thus it is an algebraic closure of F. Uniqueness of algebraic closures up to an isomorphism fixing F gives $\overline{E} \cong \overline{F}$.

Exercise 49.12: Closures of rational function fields

Since π is transcendental, $\sqrt{\pi}$ is transcendental too, and substitution $\sqrt{\pi} \mapsto x$ gives $\mathbb{Q}(\sqrt{\pi}) \cong \mathbb{Q}(x)$. Their algebraic closures are consequently isomorphic. The extension $\overline{\mathbb{Q}}(x)/\mathbb{Q}(x)$ is algebraic, so an algebraic closure of either is an algebraic closure of the other by Exercise 49.11. Finally the algebraic closure of $\mathbb{Q}(\sqrt{\pi})$ inside $\mathbb{C}$ is indeed algebraically closed: a root in $\mathbb{C}$ of a polynomial over it is algebraic over that field, hence algebraic over $\mathbb{Q}(\sqrt{\pi})$. These observations prove the requested isomorphism.

Exercise 49.13: Index is bounded by degree

For a simple algebraic extension $F(\alpha)/F$, the index is the number of distinct roots of the minimal polynomial, hence is at most its degree. Write a finite extension as a finite tower of simple extensions. Index and degree are both multiplicative along the tower, so multiplying the inequalities at each step gives $\{E : F\} \leq [E : F]$. Here braces denote the number of embeddings into a fixed algebraic closure, not a quotient field.